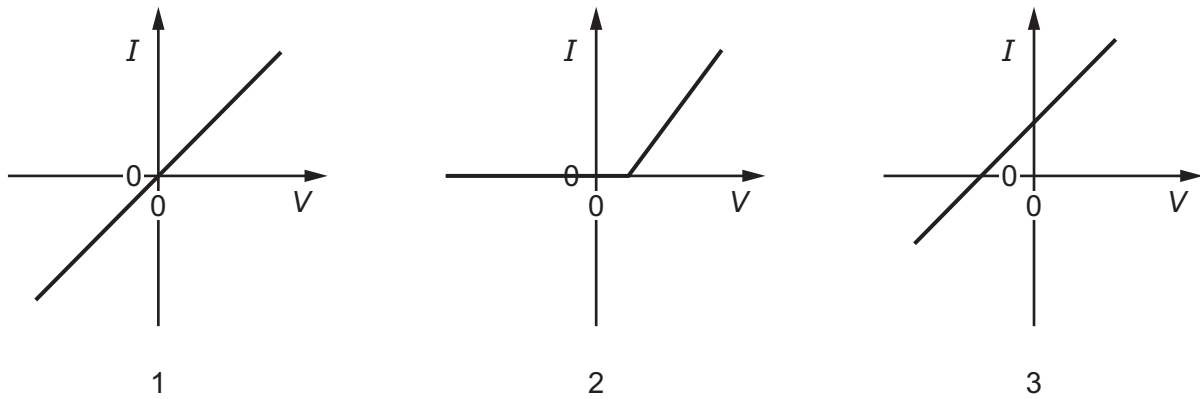


33 The I – V characteristics for three electrical components are shown.



Which components obey Ohm's law?

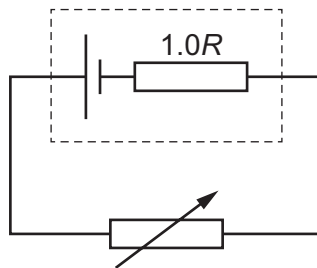
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 1 only

34 A resistor of resistance $200\ \Omega$ is connected to a supply that has a p.d. of $4.00\ \text{V}$.

How many electrons enter the resistor in $4.00\ \text{s}$?

- A** 3.13×10^{16} **B** 1.25×10^{17} **C** 5.00×10^{17} **D** 1.25×10^{21}

35 The diagram shows a cell of internal resistance $1.0R$ connected to a variable resistor.



When the variable resistor has an initial resistance of $2.0R$, the current in the cell is I_1 and the terminal potential difference (p.d.) across the cell is V_1 .

The variable resistor is now adjusted to a new resistance of $4.0R$.

What is the new current in the cell and the new terminal p.d. across the cell?

	current	terminal p.d.
A	$0.50I_1$	$1.0V_1$
B	$0.50I_1$	$1.2V_1$
C	$0.60I_1$	$1.0V_1$
D	$0.60I_1$	$1.2V_1$